MINING DEALER BEST PRACTICE

Support for transportation, R&I of 793’s front Wheel, Brake and Suspension Group

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

Support for transportation, R&I of 793’s front Wheel, Brake and Suspension Group

1.04-161215-02

1.0 Introduction

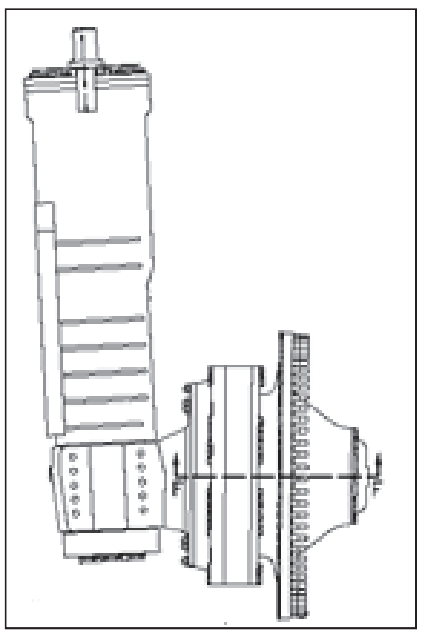
In the event of Maintenance and Repair of mining equipment, large-sized and heavyweight components and pieces must be handled. This represents a challenge for personal who must understand and apply the specialized procedures and tools in order to carry out quality work within the required time, while insuring the safety of everyone involved.

This Best Practice proposes a solution for transport, storage and R&I (Removal & Install) of the front wheel, Brake and Suspension Group of Caterpillar mining trucks model 793. The solution proposed consists of the use of two (2) supports for transport, storage and replacement of this component, which were specially designed for this purpose.

This solution aims to improve safety, efficiency, quality, and resources required in the R&I of this component and, at the same time, provide a safe system to protect the component integrity during transport and storage.

2.0 Best Practice Description

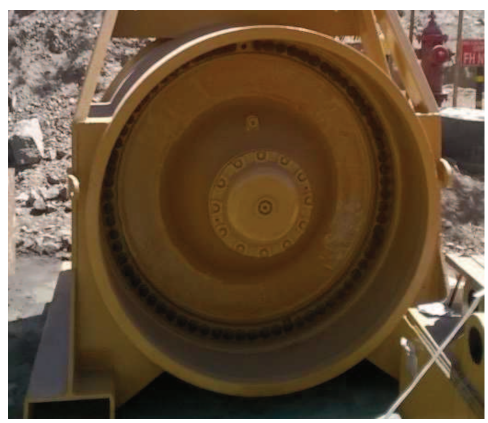
This Best Practice (BP) proposes fabricating and using specially designed supports (2) to be used both in R&I (Removal & Installation) and transportation of the front wheel assembly (WHEEL, BRAKE and SUSPENSION GP) of 793 trucks.



Pictures of the proposed supports being used during the storage of a front wheel assembly of a 793 truck.

When implementing these supports, at least two of them must be manufactured. One to be used during the removal of the wheel assembly, and the other to be used with the new component that will be installed. The same supports are used to transport and storage of the components.

In the design and manufacture of these supports, a 793 truck tire rim was used, part number 8X2642.



As mentioned above, two supports must be used to allow their proper application. One to remove the wheel assembly, and the other to install the replacement, thus avoiding additional operations of component manipulation that can be of high risk.

Installation of a front wheel assembly using the recommended supports.

Our recommendation is to use a 16 ton capacity forklift. The total weight of the front wheel assembly installed on the proposed supports is:

Weight of the 793 front wheel assembly: 5,496 kg – 12,116 lb.
Weight of the support proposed: 1,800 kg – 3,968 lb.

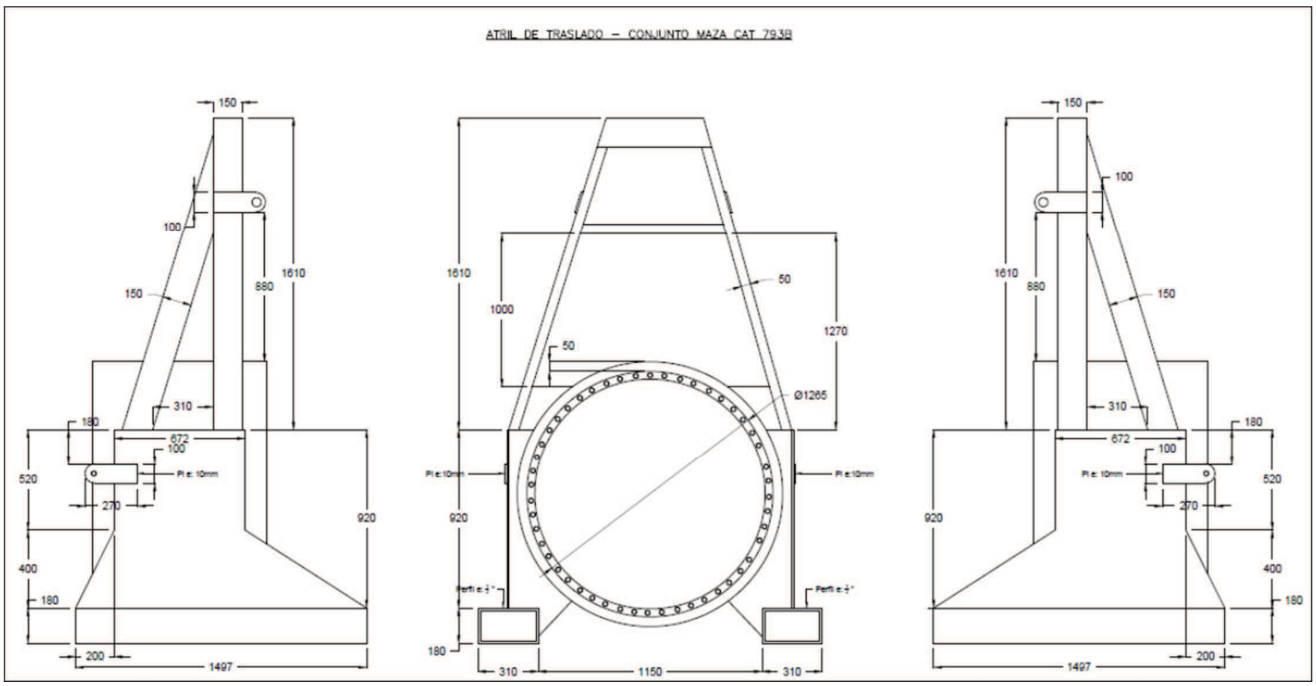
3.0 Steps for implementation

The general steps we recommend following when implementing this Best Practice are:

3.1 Manufacturing of the supports according to the schemes attached.
3.2 Technical certification
3.3 Practical tests of the supports
3.4 Final adjustments
3.5 SOP (Standard Operation Procedure) documentation
3.6 Staff training
3.7 Start-up and benefit validation

3.1 Manufacturing of the supports according to the schemes attached

The Supports were manufactured locally according to the following design specifications. This design was made by the Customer in the Finning – Candelaria contract.



Manufacture scheme attached in the Best Practice information to permit reading the dimensions.

3.2 Technical certification

Welding certification and capacity analysis of the device once it is manufactured is important. Technical certification for this type of supports is needed to ensure its safe use as well as comply with our clients' standards.

3.3 Practical tests of the supports

Before using these supports we recommend to run tests in the workshop, in a safe and controlled environment, in order to identify and make the final adjustments required.

3.4 Final adjustments

At this stage, we recommend making the necessary adjustments that may have been detected on the practical tests. In this type of supports/tools, possible adjustments should be minor, and normally related to surface, paint, and other finishes.

3.5 SOP (Standard Operation Procedure) documentation

To ensure the right operation of this device we recommend documenting the standard operation procedure (SOP).

Likewise, we recommend reviewing the local specific safety regulations, in order to adapt the SOP to your specific local regulations / needs.

3.5 Staff training

Adequate training is important for the staff who are going to use the supports regularly.

We recommend to include in this training both the standard procedure (SOP) review and practice in a safe and controlled environment.

3.7 Start-up and benefit validation

Once the supports are put into operation, there must be a follow-up to ensure the proper use and record the benefits.

Once the expected benefits are obtained, the correspondent Standard Job must be improved / updated, by the Planning and Scheduling department. Special attention must be paid to the upgrade of all related safety procedures.

4.0 Benefits

The main benefits of this Best Practice are located in five (5) areas: Safety, Efficiency, Quality, Transportation and Storage Resources and Logistics.

Safety: The use of the recommended supports contributes significantly to the safe R&I of the 793 front wheel assembly.

Efficiency and Quality: The time invested in front wheel replacement work is significantly improved (machine shutdown time and manpower time). Likewise, doing the work in a safe and controlled way, also has a positive effect on the quality of its execution. When technicians have better control of replacement maneuvers, they work better and fulfil the task in a better way and in shorter time.

Improvements in time and quality will have a positive influence on the KPI's of MTTR and MTBS respectively.

Resources: As mentioned before, the use of the proposed supports improves task execution times, thus the resources to allocate to this particular work.

Manpower improvement will have a positive influence on the maintenance ratio (MR). It is important to register adequately man hours invested in this work.

Transportation Logistics and Component Storage: The proposed supports are used both for component R&I work and component transportation and storage. This adds benefits related to the preservation of the integrity of the component as well as its manipulation and storage.

Front wheel assembly on its support during storage.

5.0 Resources required

The resources required are those specified with regard to the manufacture of the supports recommended. Remember that two (2) supports are required to the implementation of this Best Practice.

6.0 Additional support / References

The following information is attached to this Best Practice

- Manufacture scheme

7.0 Other Best Practices related

N/A

8.0 Acknowledgements

Best Practice generation, implementation, and documentation

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